

Curriculum Vitae

Personal Information

Name	Xin Wu	M/F	Male
Birth Date	Oct. 1997	Phone	13060828414
Nationality	China	Email	xinwuchn97@gmail.com; xinwu@iis.u-tokyo.ac.jp
Affiliation	IIS, UTokyo	Position	Project Assistant Professor

Education Background

2018.09 – 2023.12	Ph.D. in Solid Mechanics, South China University of Technology, China
2022.11 – 2023.10	Visiting Associate Research Fellow, The University of Tokyo, Japan
2014.09 – 2018.07	B.Eng. in Engineering Mechanics, Chang'an University, China

Work Experience

2026.04 – Present	Project Assistant Professor, Institute of Industrial Science, The University of Tokyo
2024.04 – 2026.03	JSPS Foreign Special Research Fellow, Institute of Industrial Science, The University of Tokyo

Research Interests

Phonon thermal transport; Energy materials; Solid-state battery; Density functional theory calculations; Machine-learned molecular dynamics; Data-driven rational design

Publications

In recent years, **17 SCI-indexed papers** have been published as **first/corresponding/co-first author**, with another 3 currently in preparation. In addition, 9 papers have been published as a co-author, with another 5 in preparation.

1. **Xin Wu***, Xin Huang, Lei Yang, Zhongwei Zhang, Yangyu Guo, Sebastian Volz, Qiang Han*, Masahiro Nomura*, Suppressed thermal transport in mathematically inspired 2D heterosystems, *Carbon*, 2023, 213, 118264.
2. **Xin Wu**, Zhang Wu, Ting Liang, Zheyong Fan*, Jianbin Xu, Masahiro Nomura*, Penghua Ying*, Phonon coherence and minimum thermal conductivity in disordered superlattices, *Physical Review B*, 2025, 111, 085413.
3. **Xin Wu***, Yunhui Wu, Xin Huang, Zheyong Fan, Sebastian Volz, Qiang Han*, Masahiro Nomura*, Isotope interface engineering for thermal transport suppression in cryogenic graphene, *Materials Today Physics*, 2024, 46, 101500.
4. **Xin Wu**, Qiang Han*, Phonon thermal transport across multilayer graphene/hexagonal boron nitride van der Waals heterostructures, *ACS Applied Materials & Interfaces*, 2021, 13, 32564-32578.

5. **Xin Wu***, Masahiro Nomura*, Mathematically inspired structure design in nanoscale thermal transport, *Nanoscale*, 2025, 17, 3003-3013.
6. **Xin Wu**, Qiang Han*, Transition from incoherent to coherent phonon thermal transport across graphene/h-BN van der Waals superlattices, *International Journal of Heat and Mass Transfer*, 2022, 184, 122390.
7. **Xin Wu**, Penghua Ying, Chunlei Li*, Qiang Han*, Dual effects of hetero-interfaces on phonon thermal transport across graphene/C3N lateral superlattices, *International Journal of Heat and Mass Transfer*, 2023, 201, 123643.
8. **Xin Wu**, Qiang Han*, Maximum Thermal Conductivity of Multilayer Graphene with Periodic Two-Dimensional Empty Space, *International Journal of Heat and Mass Transfer*, 2022, 191, 122829.
9. **Xin Wu**, Qiang Han*, Thermal Transport in Pristine and Defective Two-Dimensional Polyaniline (C₃N), *International Journal of Heat and Mass Transfer*, 2021, 173, 121235.
10. **Xin Wu**, Qiang Han*, Tunable anisotropic in-plane thermal transport of multi-layer graphene induced by 2D empty space: insights from interfaces. *Surfaces and Interfaces*, 2022, 33, 102296.
11. **Xin Wu**, Qiang Han*, Semidefective Graphene/h-BN In-Plane Heterostructures: Enhancing Interface Thermal Conductance by Topological Defects, *Journal of Physical Chemistry C*, 2021, 125, 2748-2760.
12. **Xin Wu**, Qiang Han*, Thermal Conductivity of Defective Graphene: An Efficient Molecular Dynamics Study Based on Graphics Processing Units, *Nanotechnology*, 2020, 31, 215708.
13. **Xin Wu**, Qiang Han*, Thermal conductivity of monolayer hexagonal boron nitride: From defective to amorphous, *Computational Materials Science*, 2020, 184, 109938.
14. **Xin Wu**, Qiang Han*, Directional Gradientless Thermoexcited Rotating System Based on Carbon Nanotubes and Graphene, *Journal of Nanomaterials*, 2019, 1-9.
15. Yuyuan Wang[#], Mengmeng Jia[#], **Xin Wu[#]**, Jiaxin Liang, Zixiang Meng, Jianshuang Wei, Tao Liu, Yuyang Yi*, Jiyun Hu*, and Jingyu Sun*, *Advanced Functional Materials*, 2026, 36, e76196
16. Chuang Hou[#], Qilong Wu[#], **Xin Wu[#]**, Jiamin Wu[#], Hirohiko Fukagawa, Manato Tateno, Yi Liu, Masahiro Nomura, Naoji Matsuhisa*, Guoan Tai*, Strain-Tunable Phonon Coupling and Polarization Optoelectronics in Supersaturation-Grown Boron Phosphide Nanowires, *Angewandte Chemie International Edition*, 2026, Accepted
17. Xiaonan Wang[#], **Xin Wu[#]**, Penghua Ying*, Zheyong Fan, Jin Zhang, and Huarui Sun*, Interface phonon modes governed ultrahigh thermal conductance of diamond/cubic boron nitride interfaces, *International Journal of Heat and Mass Transfer*, 2026, Accepted

Projects

- (1) JSPS Grants-in-Aid for Scientific Research (No. 24KF0027). Structural design-driven novel phonon thermal transport in two-dimensional structures and devices. (2 MJPY, Completed, Principal Investigator)
- (2) Japan Science and Technology Agency (JST) ALCA-Next (No. JPMJAN23E3). Development of 3DIC thermal dissipation technology based on phonon engineering. (Ongoing, Participant)
- (3) Japan Science and Technology Agency (JST) SICORP EIG CONCERT-Japan (No. JPMJSC22C6). Phonon

-
- engineering in precisely assembled atomically thin layers. (Ongoing, Participant)
- (4) National Natural Science Foundation of China (No. 11972160). Research on the propagation of surface acoustic waves based on the curving behavior of flexible electronic structures and its control mechanism. (Completed, Participant)
- (5) National Natural Science Foundation of China (No. 11902117). Acoustic metamaterials and geometric topology optimization as well as elastic wave control. (Completed, Participant)

Awards

- ◆ JSPS Foreign Special Research Fellow, Japan Society for the Promotion of Science (2024)
- ◆ National Scholarship for Doctoral Students (2022, 2021, 2020)
- ◆ President's Scholarship, South China University of Technology (2022, 2021, 2020)
- ◆ China Scholarship Council Scholarship (2023)
- ◆ First Prize in Natural Science, Guangdong Society of Theoretical and Applied Mechanics (5/6, 2024)

Academic Appointments

- Member, The Japan Society of Applied Physics
- Member, Mechanics Society of Guangdong
- Reviewer for journals including npj Computational Materials, npj 2D Materials and Applications, Material Today Physics, Advanced Composites and Hybrid Materials, Physical Letter A, International Communications in Heat and Mass Transfer, International Journal of Heat and Mass Transfer, Applied Thermal Engineering, Frontier in Physics, etc.